

MUHAMMAD HAMMAD

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Portfolio: <https://muhammad-hammad-fast.vercel.app>

PROFESSIONAL SUMMARY

Results driven Computer Science student with hands on experience building agentic AI systems, GenAI applications, and LLM-powered pipelines. Proficient in Python, FastAPI, LangChain, and cloud deployment, with a strong foundation in machine learning and full-stack development. Passionate about developing intelligent, production ready solutions that bridge cutting edge AI research and real world impact. Actively seeking opportunities to contribute to innovative teams pushing the boundaries of AI engineering.

TECHNICAL SKILLS (Data → ML/DL → LLMs → APIs → CI/CD → Cloud)

- **Languages:** Python, C, C++, SQL
- **Data & EDA:** Pandas, NumPy, Matplotlib, Seaborn data cleaning, analysis, and visualisation
- **Machine Learning:** Scikit-learn, XGBoost, Random Forest, linear and tree-based models, feature engineering, hyperparameter tuning (Optuna), model pipelines
- **Deep Learning:** TensorFlow foundational neural network architectures and training workflows
- **Generative AI / RAG:** LangChain, LangGraph, LangSmith, FAISS, ChromaDB, vector embeddings, retrieval pipelines
- **APIs & Backend:** FastAPI high-performance REST APIs for ML and LLM inference services
- **Databases:** PostgreSQL, MySQL, SQLite
- **DevOps & Cloud:** Docker, GitHub Actions (CI/CD), AWS EC2 and S3, IAM basics
- **LLMs & RL:** HuggingFace, OpenAI API, OpenAI Gymnasium
- **Tools:** Git, VS Code, Jupyter Notebooks, Streamlit

PROJECTS

Research Paper RAG: *Personal Project SaaS* FastAPI · LangChain · FAISS · Next.js · PostgreSQL · AWS

- Users uploads PDFs and interact via chatbot, structured self-explanation, and knowledge graph views. Implemented provider-agnostic LLM design supporting Grok and OpenAI APIs; containerised with Docker and deployed on AWS EC2/S3

AutoInsights: *Personal Project ML App* Tensorflow · FastAPI · Streamlit · OpenAI

- A pipeline that ingests dataset(csv), does EDA and learns features and Recommends ranked model.

Book Recommendation System: *Personal Project ML App* Scikit-learn · XGBoost · Optuna · Streamlit

- Trained and compared multiple ML models with Lasso feature selection and Optuna hyperparameter tuning; deployed to Streamlit Cloud

ArguMind: *University Project AI* Python · Grok · Gemini Next.js · FastAPI · LangChain · Langgraph

- 4 llm agents debate over an argument to comeup with a solution

Packet Routing with Reinforcement Learning: *University Project AI* Python · OpenAI Gymnasium

- Designed an RL agent for adaptive packet routing across network graphs, benchmarking against shortest-path algorithms

Sportics Management System: *University Project Database Management* PostgreSQL · SQL

- Full database-backed system to manage society events from drafting through scoring; designed complex relational schemas and stored procedures

Additional ML & AI Projects

- **Classification:** ChurnRatePredictor, PersonalityPredictor, DiabetesPredictor end-to-end supervised pipelines
- **Recommendation & NLP:** MusicRecommender, SpamDetector, content-based filtering, and text classification
- **Deep Learning:** ImgClassifier (CNN), NextWordPredictor (LSTM)
- **Generative AI:** ChatBots and ResearchPaperRAG LangChain conversational agents with retrieval-augmented generation

EDUCATION:

Intermediate (Pre-Engineering) 2021 – 2023 Fauji Foundation College, Tando M.Khan 83%

BS Computer Science 4th year@ FAST NUCES, Karachi CGPA: 3.0

- Relevant coursework: Data Structures, OOP, Operating Systems, Computer Networks, Databases, AI, Probability & Statistics, Computer Architecture
- Activities: Coders Cup, Procom, DevDay, Probattle inter and intra-university competitions

CAREER FOCUS

Aspiring AI/ML Engineer with hands-on experience in agentic systems, GenAI applications, and full-stack AI deployment. Eager to contribute to innovative teams building real-world intelligent solutions.